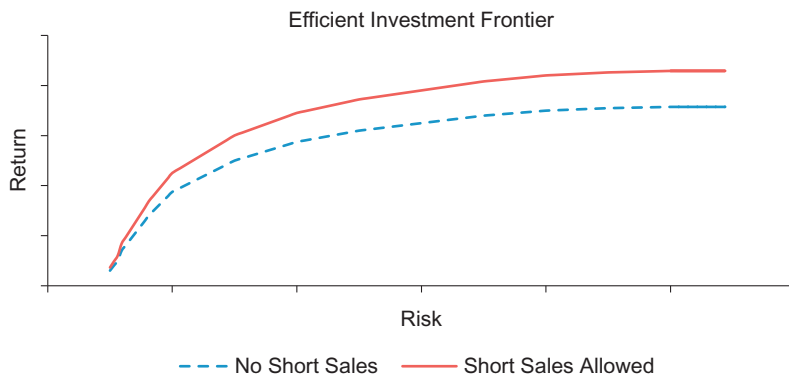


Example: Efficient Trading Frontier w/ and w/o Short Positions

In this example, we construct the efficient frontier using the techniques above for a subset of 100 large cap stocks from the SP500 index. We construct these portfolios with and without a short sales constraint. That is, one portfolio is a long-only portfolio where all weights have to be positive, and the other portfolio contains both long and short positions and the weights could be positive or negative. This is shown in [Figure 10.1](#). Notice how the efficient frontier with short sales allowed provides higher returns for the same risk than the no short sales case. This is because investors can use short positions to better manage risk in the portfolio. Additionally, this example demonstrates how the usage of constraints (in this example no short sales allowed) may result in reduced portfolio performance.

Example: Maximizing Investor Utility

In this example, we show how investors determine their investment portfolio based on their utility preferences. Utility preferences, expressed in terms of indifference curves, are the set of all return-risk portfolios that provide the investor with equal quantities of “happiness.” Investors, therefore, are indifferent to which portfolio on the indifference curve they actually own since all of these portfolios provide the same quantity of economic utility. Investors, of course, will always prefer higher returns for the same level of risk than lower returns. Thus, the goal of the investor is always to be on the highest indifference curve possible.



■ Figure 10.1 Efficient Trading Frontier.